

LED Scrolling Display

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Most outdoor LED displays and some indoor LED displays are built around individually-mounted LEDs. A cluster of red, green and blue diodes is driven together to form a full-colour display.

In a dot-matrix LED display, the LEDs are wired together in rows and columns to minimise the number of

pins required to drive them. For example, an 8x8 matrix of LEDs, shown in Fig. 1, would need 64 I/O pins—one for each LED pixel. By wiring all anodes together in rows (R1 through R8) and cathodes in columns (C1 through C8), the required number of I/O pins is reduced to 16. Each LED is addressed by its row and column number. In Fig. 1, if R4 is pulled high and C3 is pulled low, the LED in the fourth row and third

column will turn on. Alphabets and numerals can be displayed by fast scanning of either rows or columns. In this project, column scanning has been used.

Working of a dot-matrix display

Fig. 2 shows which LEDs in a 5x7 matrix of LEDs are to be turned on to display the English alphabet A. The seven

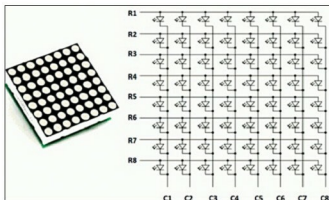


Fig. 1: Structure of an 8x8 LED dot-matrix display

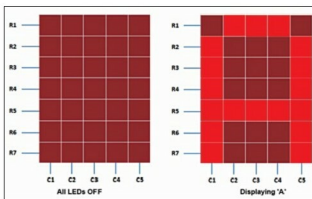


Fig. 2: 5x7 array of LEDs

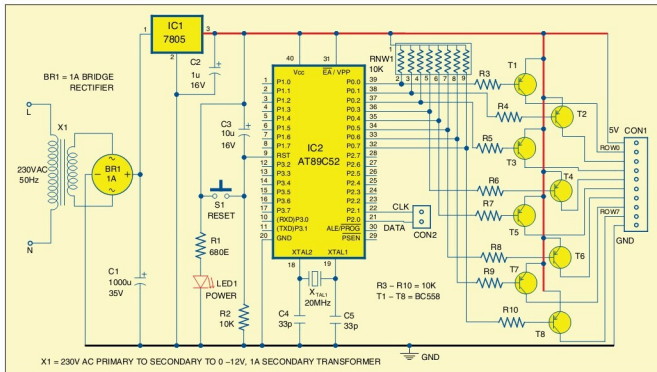


Fig. 3: Power supply and controller circuit